Slowing down Early Stage Model Collapse in Language Models through Diversity-Enhancing Decoding

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Abstract

This project examines whether decoding strategies affect early-stage model collapse in recursive Chinese text generation. Using Qwen1.5-0.5B fine-tuned on Chinese Wikipedia data, we compare greedy search, temperature sampling, top-p sampling, and Diverse Beam Search (DBS). Results show severe repetition under greedy decoding, strongest overall stability under temperature sampling, moderate performance under top-p, and sequence-level diversity but internal repetition gains under DBS.

1 Introduction

1.1 Motivation, Aim and Research Question

Large language models are no longer separate from the text environment in which they are trained. Their outputs now circulate in corpora such as web pages, forum posts, summaries, question-answering data, and synthetic datasets, making it increasingly likely that machine-generated text will be collected again as training data. This creates a practical risk: future models may learn not only from human-written language, but also from the repeated outputs of earlier models. Recent work describes this process as model collapse, where recursive training on generated data can gradually distort the model's approximation of the original data distribution.

This project does not attempt to model full-scale collapse in large pretraining pipelines. Instead, it focuses on an earlier and more controllable stage of the problem: whether synthetic text already shows signs of degeneration when it is recursively generated and reused for fine-tuning. This matters because synthetic text is produced by a particular model under a particular generation procedure. If that procedure repeatedly favours conservative or formulaic continuations, the resulting data may already represent a narrowed version of the original language distribution.

The project investigates this issue in a small-scale Chinese text-continuation experiment using Qwen1.5-0.5B. This setting extends existing work on decoding and model collapse, which has mainly been evaluated on English corpora such as WikiText-2 and with different model families. The project also compares token-level decoding methods with a sequence-level diversity method, asking whether these strategies affect early-stage degeneration in different ways. The central research question is therefore whether different decoding strategies lead to different diversity and repetition patterns in recursively generated Chinese synthetic text. The scope of the project is deliberately limited. It does not claim to solve model collapse, nor to reproduce the full dynamics of web-scale recursive pretraining. Instead, it examines early collapse-like symptoms in a controlled continuation setting, leaving the detailed model, data construction, decoding methods, and evaluation metrics to the following sections.

1.2 Related Work

Research on model collapse provides the starting point for this project. Shumailov et al. (2024) formalise model collapse as the degradation that occurs when generative models are recursively trained on data produced by earlier models. They show that repeated training on generated data can damage the model's approximation of the original data distribution, especially in low-probability regions. For language modelling, the risk is that recursive training may gradually remove rare expressions, unusual facts, and stylistic variation from the original distribution. Synthetic text therefore functions not only as extra data, but as a distribution that later models learn from.

Recent work has connected this problem more directly to large language models and to the way synthetic data are generated. Drayson et al. (2025) investigate how decoding strategies affect model collapse in recursive LLM training. Their experiments use English WikiText-2 data and evaluate several decoding strategies, including greedy decoding, beam search, pure sampling, temperature sampling, top-k sampling, and nucleus sampling. They assess collapse through model performance, generated text quality, and similarity to human text, and show that decoding choices can lead to different degrees of degradation. This is the closest prior work to the present project, but our setting differs in two ways: we focus on Chinese text generated by Qwen1.5-0.5B, and we include a sequence-level diversity method, Diverse Beam Search, alongside token-level stochastic methods.

Another line of work examines whether collapse can be reduced by changing the composition of the training data. Shumailov et al. (2024) show that retaining a proportion of human-written data can mitigate collapse, while Gerstgrasser et al. (2024) argue that accumulating real data together with synthetic data can avoid some recursive degradation. These approaches are important, but they mainly address the mixture of human and synthetic data used for training. The present project instead looks at an earlier stage: how the synthetic portion is generated before it becomes training data.

The work on neural text degeneration explains why this generation stage matters. Holtzman et al. (2019) show that maximisation-based decoding can produce bland or repetitive open-ended text, and propose nucleus sampling as a way to reduce this problem. Wiher et al. (2022) further show that decoding methods involve task-specific quality-diversity trade-offs, so their effects need to be tested in the target setting. Sequence-level diversity has also been studied through methods such as Diverse Beam Search, which encourages different candidate sequences to explore different regions of the output space Vijayakumar et al. (2016). Recent work on structure-conditional decoding similarly shows that open-ended outputs vary not only in token choice, but also in structure and meaning Eikema et al. (2025). This motivates our comparison of token-level and sequence-level decoding methods, which may affect early-stage degeneration in different ways.

2 Experimental setup

2.1 Model

We use Qwen1.5-0.5B, a lightweight open-source language model pretrained on large-scale multilingual corpora, including Chinese data (Bai et al., 2023). Its small parameter size makes repeated fine-tuning and generation computationally feasible, while its Chinese generation capability makes it suitable for the Wikipedia continuation task.

The Qwen model we select is used for word segmentation, model fine-tuning, and text generation stages.

2.2 Data

For the data set, we use Chinese Wikipedia data set (Wikimedia Foundation, 2026). In total, 30,000 files are randomly selected from the whole data set. We extract a random paragraph of each file and retain it if the paragraph length exceeds 128 characters. Then, Qwen 1.5-0.5B tokenizer is used to segment the text into words. Paragraphs with more than 128 tokens (words) are retained and truncated to a length of 128. We randomly choose 1,400 paragraphs; 1,000 of them are used as the training set, 200 as the validation set, and 200 as the test set.

The validation and test sets remain unchanged throughout the experiment and are used only for evaluation. The training set is used to fine-tune the baseline model (round 0). The first 64 tokens of each paragraph in the training set are set as a fixed prompt for model generation. In each subsequent round, each paragraph in the training set is composed of the first 64 tokens of the initial training set (human data) plus the subsequent 64 tokens generated by the model of previous round (synthetic data).

Thus, for the initial round of model fine-tuning, we use human data. During subsequent rounds of model fine-tuning, we use partial synthetic data.

3 Methods

3.1 Procedure

Our methods and procedures refer to the research of Shumailov et al. (2024) and Drayson et al. (2025). We iteratively conduct the following procedure for 5 rounds. In each iteration, we go through the process of model fine-tuning, model generation, and the evaluation of model generated text.

For model fine-tuning, we fine-tune the baseline model (Qwen 1.5-0.5B) using the training data for each round, and evaluate model performance each round on validation and test sets by computing perplexity. LoRA (Hu et al., 2022) is applied for fine-tuning and performs better on this small dataset compared to full fine-tuning. It is suitable for this task in that it enables lightweight adaptation without updating the entire model. Moreover, it can reduce the risk of model overfitting, and can get more stable results while fine-tuning on limited data as well.

For the generation task, we choose a text continuation task. The first 64 tokens of each paragraph in the human data are set as the fixed prompt and the model is required to generate the subsequent 64 tokens. We select several decoding algorithms. At each iteration, a specific decoding algorithm is applied to generate the subsequent sequences of tokens. The prompt is fixed in order to better compare the generated text across different rounds and different decoding methods. We also define diversity metrics to evaluate the quality of model generated text in each round.

In round 0, we fine-tune the baseline model (Qwen 1.5-0.5B) using Chinese Wikipedia dataset (human data). Then, the model after fine-tuning will be used to generate the training data for the next round. Since the model generated the subsequent 64 tokens of the prompt (first 64 tokens of human data), for the following rounds, the training data is partially synthetic, consisting of the prompt and the subsequent generated sequence of tokens. Partial synthetic data are used to make it more in line with the actual situation of real-world corpora.

In the subsequent rounds, we iteratively fine-tune the baseline model using the partial synthetic data generated by the previous model and iteratively generate synthetic training data adopting the latest model. During this process, early stage of model collapse is simulated, as it is repeatedly fine-tuned on synthetic data generated by itself.

3.2 Decoding algorithms

When the early signs of model collapse occurs, the diversity of text generated by the model reduces. To test our hypothesis that different diversity-enhancing decoding strategies would slow down the process of model collapsing and enhance the diversity of the generated text, we select the following 4 diversity-enhancing decoding strategies (referring to Wiher et al. 2022).

- **Baseline**: Greedy Search
- **Token-level Methods**: Temperature Sampling, Nucleus (Top-p) Sampling
- **Sequence-level Methods**: Diverse Beam Search (Vijayakumar et al., 2016)

Greedy search decoding generates the next token with the highest probability each time. It is set as the baseline as this decoding algorithm doesn't encourage diversity. Thus, by comparing with the baseline condition, we can assess the performances of these diversity-enhancing decoding algorithms.

We select both token-level and sequence-level decoding algorithms to test our second hypothesis that the latter would outperform the former in slowing down early signs of model collapse and increase context diversity. All three decoding algorithms are diversity-enhancing strategies.

Two token-level decoding algorithms, temperature sampling and nucleus (top-p) sampling, are chosen. **Temperature sampling** could add randomness to the model's generation. While generating the next token, it introduces a temperature parameter before softmax to control the "sharpness" of the probability distribution. If the distribution is less sharper, the model is less likely to select high-probability tokens. Thus, the randomness is increased.

Nucleus (top-p) sampling adjusts the size of the set of the candidate next tokens so as to increase diversity while generating. It remains the smallest candidate set of tokens, the cumulative probability mass of which does not exceed the threshold p. The probabilities are then renormalised and the next token is selected according to the new distribution.

For sequence-level decoding algorithm, we choose **diverse beam search** (DBS). DBS extends standard beam search by promoting diversity among beam groups. It splits the beam into multiple groups and each group performs beam search while punishing sequences similar to those generated by the previous groups or repetition. Thus, it reduces the probability that different beams may provide the same generation. In this case, DBS could balance high likelihood with diversity, producing high-quality but distinct sequences.

3.3 Evaluation metrics

3.3.1 Metrics for model performance

To evaluate model performance each round, we fine-tune the model on the training set and test it using validation and test sets. The loss and **perplexity** of the validation and test sets are calculated to measure performance. It evaluates how certain the model is while predicting the next token (Jurafsky and Martin, 2026). The lower the perplexity, the more confident and accurate the model is while predicting.

3.3.2 Metrics for text generation quality

Diversity metrics are defined to evaluate the quality of text generated by the model of each round. The following three metrics are adopted.

Dist-n: Dist-n calculates the number of distinct n-grams divided by the total number of n-grams (Li et al., 2016). The higher the Dist-n is, the greater token-level diversity of the text will be. In this case, we calculate dist-1 and dist-2.

Self-BLEU: Self-BLEU is the average BLEU score of the document, which measures how similar each generated output is to the others (Zhu et al., 2018). The lower the Self-BLEU score is, the greater sequence-level diversity of the text will be.

Severe repetition rate: Severe repetition rate is calculated as the number of samples with severe repetition divided by the total number of samples (Holtzman et al., 2019; Wiher et al., 2022). Severe repetition is defined as when a phrase (minimum length of 2 tokens) occurs three times or more in a sequence. The higher the repetition rate is, the lower the diversity is within sequences. We don't restrict the repetition as the phrase repeated continuously. In this case, some repetition of the sentence patterns or structures can be detected. Thus, the severe repetition rate would be relatively higher.

3.4 Replicability

The data and code of this article are in the attachment file, and the code is run in Google Colab using the T4 GPU. The data is a subset of Chinese Wikipedia dataset, which is available at `https://dumps.wikimedia.org/zhwiki/`. The Qwen model is available at `https://huggingface.co/Qwen/Qwen1.5-0.5B?utm_source=chatgpt.com`. Hence, the research can be replicated under the same hardware and software environment.

4 Results and analysis

4.1 Simulating early stage of model collapse

Throughout the recursive training procedure, the loss and perplexity of the training set, starting from the second round where partial synthetic data are used, continue to decrease, while the loss and perplexity of the validation and test sets, which use human data, fluctuate and generally increase. This indicates that the model learns the synthetic data distribution, while its distribution gradually deviates from the real data distribution. Moreover, Dist-1 and Dist-2 of the text generated by the model are lower than those of the original training data (Dist-1 = 0.146, Dist-2 = 0.753). This reveals that token diversity has decreased and the generated distribution has narrowed. The severe repetition rate is also much higher than that of the original data (0.360). These are early signs of model collapse.

Nevertheless, the model does not collapse immediately. In some rounds, perplexity decreases before increasing again. For instance, from round 1 to round 2 of DBS decoding, the perplexity of the test set drops from 28.819 to 28.708. This may be interpreted as a weak self-distillation effect, where the model temporarily adapts to its own generated text and reduces some prediction errors (Shenfeld et al., 2026). However, as the synthetic data distribution moves further away from the origin human data, the overall trend still points toward early-stage degeneration.

4.2 Model performance on validation and test sets

Perplexity is used to measure model performance after fine-tuning in each round, indicating whether the model's generalisation ability changes across recursive training. According to Figure 1, the validation and test perplexities increase after five rounds for all four decoding algorithms, while the perplexity of the training set keeps decreasing. This suggests a degradation in the model's generalisation ability. A distribution shift may occur as the model is fine-tuned on partially synthetic data but evaluated on human validation and test data during the recursive training process.

Among the four decoding algorithms, temperature sampling outperforms the other three and maintains relatively lower validation (increase from 28.649 to 29.550) and test perplexities (increase from 27.440 to 28.347). Top p sampling has the second lowest perplexities (validation: 29.979, test: 28.745), followed by DBS (validation: 30.088, test: 28.840). The baseline condition (greedy decoding) shows the strongest early collapse-like symptoms among the four strategies, with the highest final validation and test perplexities (validation: 30.340, test: 29.082).

4.3 Results of diversity metrics

4.3.1 Dist-n

Dist-n reflects the token-level diversity. According to Figure 2 (two figures on top) , temperature sampling performs the best in Dist-1 and Dist-2, with Dist-1 reducing from 0.125 to 0.121 and Dist-2 reducing from 0.683 to 0.660 respectively, close to Dist-1 (0.146) and Dist-2 (0.753) of the original training data.

Note that the Dist-1 value of the human data is still relatively small. This is partly because the experiment uses Chinese Wikipedia data, whose content is highly structured and often repeats named entities, topic terms, function words, and punctuation. In addition, the metric is computed over Qwen token IDs rather than manually segmented or semantically disambiguated words. Chinese also contains surface-identical forms that may carry different meanings in different contexts, but Dist-1 treats them as the same token. Therefore, the absolute Dist-1 values should be interpreted cautiously, and the relative trends across decoding strategies are more informative for the analysis.

Top-p sampling gets the second highest Dist-1 and Dist-2 after 5 rounds, with Dist-1 reducing from 0.118 to 0.113 and Dist-2 reducing from 0.642 to 0.595. DBS ranks the third in token-level diversity. Dist-1 of text generated using DBS increases from 0.082 to 0.105 during the five rounds, and Dist-2 increases from 0.338 to 0.430. Text generated by greedy decoding has the lowest token-level diversity. Its Dist-1 increases from 0.084 to 0.095, and Dist-2 decreases from round 0 to 1 (0.369 to 0.332) and then increases in the following rounds to 0.369.

4.3.2 Self-BLEU

Self-BLEU reflects the sequence-level diversity of the text. As shown in Figure 2 (bottom left), the text generated by DBS gets the highest self-BLEU score in the first round, and then gradually decreases in the subsequent rounds to 0.103, which eventually becomes the lowest among the four decoding algorithms and lower than the self-BLEU score of the original training set (0.120). The baseline condition gets the second highest score of 0.199 in the beginning, and reduces to 0.116, higher than that of DBS in the last round.

Self-BLEU of temperature sampling is the lowest in round 0 (0.142). Its value fluctuates and rises to 0.149 at last. Top-p sampling has the highest self-BLEU score. It increases from 0.165 to 0.175 and decreases back to 0.165 in the end.

4.3.3 Severe repetition rate

The severe repetition rate shows the degree of repetition of phrases within a sequence of generated text. The severe repetition rate of the original training data is 0.360. According to the Figure 2 (bottom right), the baseline condition has the highest severe repetition rate of 0.950 in round 0 and decreases to 0.854. Text generated using temperature sampling has the lowest severe repetition rate, with a value of 0.470 in the first round and increasing to 0.614 in the end.

The severe repetition rate of top-p sampling is the second lowest in the first round, but increases to 0.775 in the following rounds, higher than the final score of DBS. DBS, on the contrary, gets the second highest severe repetition rate in round 0, but decreases to 0.753 at last.

4.4 Comparison and analysis

4.4.1 Greedy decoding

According to the results, the baseline condition, greedy decoding, shows the strongest early-stage degeneration among the four strategies. It has the highest validation and test perplexities, indicating the weakest generalisation to the fixed human validation and test sets. In terms of generation quality, it also has the lowest Dist-1 and Dist-2 scores and the highest severe repetition rate. These results suggest that greedy decoding produces a narrow and highly repetitive synthetic distribution from the first generation round.

Although Dist-1 increases, and Self-BLEU and severe repetition rate decrease during the five rounds, this should not be interpreted as genuine recovery. Greedy decoding already produces highly degenerate text in the first round, so later improvements partly reflect the model becoming more adapted to its own simplified synthetic distribution. In other words, the model may fit the recursive training data more easily while moving further away from the actual distribution of the text (original Wikipedia data).

4.4.2 Temperature sampling

Temperature sampling shows the most stable performance in slowing down the process of early stage model collapse, as well as in maintaining generation diversity. It has the highest Dist-1 and Dist-2 among the four strategies, which is close to that of

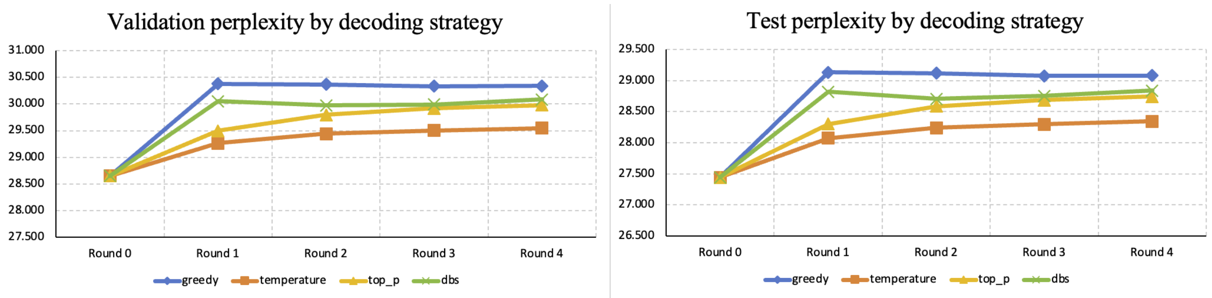

Figure 1: Validation and test perplexity across 5 recursive fine-tuning rounds

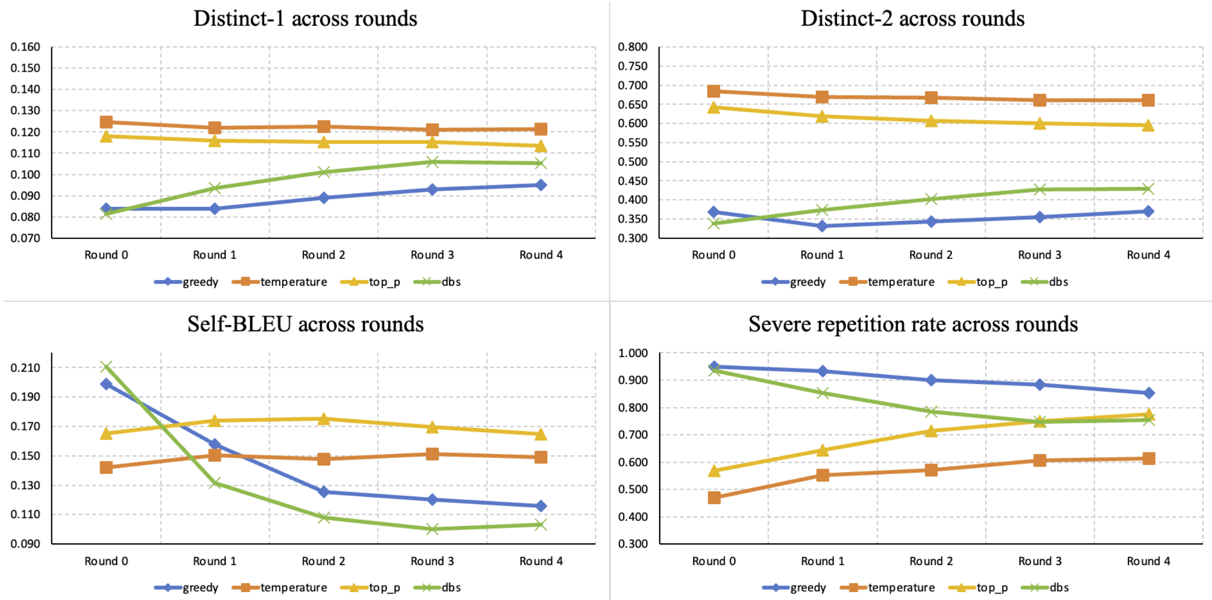

Figure 2: Evaluation Metrics Across 5 Recursive Rounds

the original human data. This suggests that temperature sampling is more effective in maintaining token-level diversity in the text. As it keeps the distribution coverage of the original human data and offers chances for the tokens with lower probability to be selected, the distribution of synthetic data generated by the model using temperature sampling has the least deviation from the real data distribution.

Although its severe repetition rate and validation/test perplexities still increase across the five rounds, they remain the lowest among the four decoding strategies. This suggests that temperature sampling does not fully prevent early-stage degeneration, but it slows the process more effectively than the other strategies. However, its final Self-BLEU score is the second highest among the four decoding strategies, indicating that temperature sampling does not produce the most diverse outputs across samples. This may be because its generated continuations remain relatively close to the structured style of Chinese Wikipedia, so some outputs still share similar patterns. Nevertheless, temperature sampling provides the best overall balance between model-level performance, token-level diversity, and within-continuation repetition.

4.4.3 DBS

DBS performs best in maintaining diversity at the sentence level, but is not good at dealing with token-level repetitions within a sequence. Since it encourages beam diversity, it has the lowest self-BLEU score of 0.103 after 5 rounds, even smaller than that of the real-world data (0.120). The sequences it generates are less similar to each other.

However, it has very poor token-level diversity, with a low Dist-1 and Dist-2 and a high severe repetition rate. The situation happens because DBS still uses beam search to generate each sequence. Despite that it has a diversity penalty, it is still a high-probability sequence search. Thus, it is difficult for DBS to prevent high-probability words from being repeatedly selected within a sequence.

It's noticed that self-BLEU and severe repetition rate of DBS start with large values in the initial round and gradually decrease in subsequent rounds. In the first round, model generation using DBS might produce "noises", especially when fine-tuning on a small dataset with relatively less diverse vocabulary. Thus, repeated phrases are more likely to occur in the sequence. In the subsequent rounds, the model learns the beam outputs of DBS, and generates more stable and less repetitive results.

Hence, in this task, using DBS as the decoding method results in a relatively strong sign of early stage model collapse and does not effectively slow down the process or maintain content diversity. However, it does increase the sequence-level diversity of the text. In order to compare the effects of these decoding algorithms on slowing down the process of model collapse, the repetition penalty or repeated n-gram penalty is not set for DBS. If these values are set, DBS may perform better in maintaining token-level diversity and the severe repetition rate of may decrease.

4.4.4 Top-p sampling

Top-p sampling presents moderate performance in slowing down early stage of model collapse and maintaining a balanced generation diversity, with the second highest Dist-1 and Dist-2 and a second highest severe repetition rate in round 0. However, the severe repetition rate increases in the recursive rounds and reaches a value (0.775) higher than the final value of DBS (0.753). Since top-p sampling remains candidate tokens, the cumulative probability mass of which does not exceed the threshold p (0.9), it is tail truncation. Thus, tokens with low probability in the tail will not be retained. As the recursive procedure goes on, the distribution of model generated data gradually shifts from the real distribution. The distribution narrows during the process as well.

Moreover, the self-BLEU score of top-p sampling is the highest (0.165) among the four, indicating a sentence-level concentration in the text generated using this decoding algorithm. It indicates that top-p sampling maintains some token-level diversity but does not effectively improve the sentence-level diversity, as it selects sentence structures with higher probabilities.

4.4.5 Summary

To sum up, fine-tuning the model recursively on the Chinese text generated by itself, different decoding algorithms do influence the extent to which model collapses, as well as the diversity of the outputs. This is in line with the study of Drayson et al. (2025) in the English text.

Among the 4 decoding algorithms chosen, temperature sampling performs best in slowing down the early stage model collapse and in maintaining token-level diversity in this task. Top-p sampling has moderate performance in maintaining token-level diversity but performs worse in maintaining sequence-level diversity. DBS performs well in maintaining sequence-level diversity but has a poor performance in maintaining token-level diversity. All three diversity-enhancing decoding algorithms help slow down early stage model collapse to some extent compared to the baseline condition. This supports our first hypothesis.

Contrary to our second hypothesis, the sentence-level decoding algorithm (DBS) doesn't outperform token-level decoding algorithms in slowing down the process of model collapse. Nevertheless, it has the advantage of maintaining sequence-level diversity, which is a limitation of token-level decoding algorithms. If combined with methods to reduce token-level repetition within a sequence while generating, sentence-level decoding algorithm has the potential to perform better in dealing with the problem.

5 Conclusions

This project examines how decoding strategies affect early-stage degeneration in recursive Chinese text generation. Using Qwen1.5-0.5B fine-tuned on Chinese Wikipedia data, we compare greedy search, temperature sampling, top-p sampling, and Diverse Beam Search across five recursive rounds. The results show that decoding choice has a clear effect on both model-level performance and generation-level quality.

Greedy decoding produces severe repetition from the first round, making it a weak high-probability reference condition rather than a useful strategy to preserve synthetic data quality. Temperature sampling performs best overall, maintaining the strongest token-level diversity, the lowest severe repetition rate, and the lowest validation and test perplexities after five rounds. Top-p sampling shows moderate performance, but its repetition rate increases more clearly across rounds. DBS achieves the lowest final Self-BLEU score, suggesting stronger diversity among generated outputs, but it doesn't effectively reduce repetition within individual continuations.

Overall, the findings suggest that diversity-oriented decoding can reduce some early collapse-like symptoms, but not all diversity methods can address the same problem. Token-level stochastic decoding is more effective for reducing within-continuation repetition, while sequence-level diversity may require additional repetition-control mechanisms to improve synthetic training data quality.

Limitations

First, the experiment is small-scale in both model and fine-tuning setup. We use only Qwen1.5-0.5B and apply LoRA fine-tuning rather than full fine-tuning or pretraining from scratch. This makes the recursive experiment computationally feasible, but it also limits how far the results can be generalised. Since LoRA updates only a small number of additional parameters, the model may respond differently to recursively generated data than it would under full fine-tuning. The findings should therefore be interpreted as early-stage degeneration under a parameter-efficient adaptation setting, rather than as a full account of model collapse in large-scale training.

Second, the data setting is limited. The experiment uses a subset of Chinese Wikipedia paragraphs, which provides a relatively clean and structured source of Chinese text but may not represent the diversity of real Chinese web corpora. Chinese Wikipedia may also contain missing words, formatting noise, encoding artefacts, or simplified–traditional conversion issues. Its relatively formulaic style may affect diversity metrics and make repeated structures more likely than in less structured corpora.

Third, the recursive training setup is simplified. Each example keeps the first 64 tokens of the original human text as a fixed prompt, while the following 64 tokens are replaced by model-generated continuations in later rounds. This partial-synthetic design keeps the prompt comparable across rounds and decoding strategies, but it does not fully reproduce how synthetic text may enter real training corpora, where prompts, continuations, document boundaries, and sources may all be generated, filtered, edited, or mixed. The experiment also runs for only five recursive rounds, which is enough to observe early degeneration patterns but not enough to support strong claims about long-term collapse dynamics.

Fourth, the evaluation metrics capture only part of generation quality. Dist-1 and Dist-2 measure lexical diversity, Self-BLEU measures similarity across generated samples, and severe repetition rate measures phrase-level repetition within individual outputs. These metrics are useful for detecting surface-level degeneration, but they do not directly measure factuality, coherence, fluency, syntactic diversity, semantic diversity, or usefulness. Severe repetition is also sensitive to tokenisation choices. Since it is computed over Qwen token IDs and counts repeated phrases of at least two tokens, it may be affected by Chinese tokenisation patterns, frequent function words, punctuation, or repeated Wikipedia-style structures. It should therefore be interpreted together with Dist-n, Self-BLEU, and qualitative inspection.

Fifth, the decoding comparison is not exhaustive. We compare greedy search, temperature sampling, top-p sampling, and Diverse Beam Search, but the results depend on the chosen hyperparameters. Different temperature values, top-p thresholds, beam sizes, beam groups, or diversity penalties could change the outcome. In particular, DBS encourages diversity among candidate sequences, but our pipeline keeps only one continuation per prompt, so this sequence-level diversity may not fully enter the final synthetic training data. We also do not use repetition penalties or no-repeat n-gram constraints, because the aim is to compare decoding strategies without extra repetition-control mechanisms.

Finally, this project studies decoding as one factor shaping synthetic data, not as a complete mitigation strategy. Future work could test larger models, longer recursive chains, broader Chinese corpora, full fine-tuning, human evaluation, semantic and syntactic diversity metrics, and combinations of decoding with filtering, repetition penalties, or human-data mixing.

Use of AI tools

Chat GPT is used to assist with coding. AI assistant of Overleaf is used to assist with grammar and spelling.

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Appendix

A Hyperparameters

A.1 Model fine-tuning

In our experiment, we conduct LoRA fine-tuning using a batch size of 4 and epochs of 3. Learning rate is set to 2e-5. The rank of LoRA (r) is set to 8, and LoRA scaling factor (alpha) is set to 16. LoRA dropout rate is set to 0.1. `q_proj` and `v_proj` are used to train the adapter. This setting is done to reduce the possibility of model overfitting or preventing overly strong LoRA update amplitude.

A.2 Decoding algorithms

For temperature sampling, we set the temperature parameter to 1.0. For top-p sampling, we set the threshold p to 0.9. For DBS, we set `num_beams` = 6, `num_beam_groups` = 3, and `diversity_penalty` = 1.0. Moreover, we use reranking to select the final output for each sequence.

B Supplementary Files

The full table of model performance and generation-level diversity metrics is provided in the supplementary file `metrics.xlsx`. This file includes the validation/test perplexity scores and generation-level metrics across recursive rounds for each decoding strategy.

The qualitative generation examples are provided in `examples.txt`. This file contains the original Chinese generated outputs across five rounds for each decoding strategy, together with English translations for readability.